

CURRICULUM VITAE

Gracious Y.B. Donkor

EDUCATION

University of Illinois at Urbana–Champaign (UIUC)

Aug 2023 – present

M.S. Microbiology

Ph.D. Microbiology

Thesis advisor: James M. Slauch, Ph.D.

Illinois State University (ISU)

Jan 2021 – July 2023

M.S. Biotechnology

Thesis advisor: Jan–Ulrik Dahl, Ph.D.

Kwame Nkrumah University of Science and Technology (KNUST)

Sept. 2013 – 2017

BSc. Medical Laboratory Technology

Thesis advisor: Alex–Owusu Ofori, Ph.D.

PUBLICATIONS/MANUSCRIPTS

1. Cakar Fatih, **Donkor G.Y.**, Abdulla S. Z., Golubeva Y., Vanderpool C. K., Slauch J. (2026) sRNA-mediated regulation of Rho action at the 3' untranslated region controls *hilD* mRNA stability in *Salmonella*. [*In preparation*].
2. **Donkor, G. Y.**, & Dahl, J. U. (2026). Implications of Bacterial Exposure to Reactive Oxygen and Chlorine Species. [*Accepted*].
3. Grant NA, **Donkor GY**, Sontz J, Soto W, Waters CM.2025. Deployment of a *Vibrio cholerae* ordered transposon mutant library in a quorum–competent genetic background. mBio.
4. Emmanuel P. Oladokun, **Gracious Y. Donkor**, Julius K. Narh, Grady D. Jacobson, Cade Ward, Patrick O. Tawiah, Lisa K. Polzer, Kevin A. Edwards, Kyle A. Floyd, Jan–Ulrik Dahl. The Synergy between a Silver–Ruthenium Antimicrobial and aminoglycosides is based on severe macromolecular damage. [*In review*]
5. Canchola, J.; **Donkor, G.Y.**; Tawiah, P.O.; Fasawe, A.; Ayim, E.; Engelke, M.F.; Dahl, J.–U. Alkyl Pyridinol Compounds Exhibit Antimicrobial Effects against Gram–Positive Bacteria. *Antibiotics* 2024, 13, 897.
6. **Donkor GY**, Anderson GM, Stadler M, Tawiah PO, Orellano CD, Edwards KA, Dahl J–U. A novel ruthenium–silver-based antimicrobial potentiates aminoglycoside activity against *Pseudomonas aeruginosa*. mSphere. 2023 Oct 24;8(5):e0019023.
7. **Donkor GY**, Dontoh E, Owusu–Ofori A (2019) A cross–sectional study on the prevalence of antibiotic use prior to laboratory tests at two Ghanaian hospitals. PLOS ONE 2019

AWARDS & HONORS

May 2025	Mame Shiao Debbie Award, UIUC
May 2025	James Beck Fellowship, UIUC
Sept 2024	James L. Fisher Outstanding Thesis Award (2023/2024), ISU
May 2024	School of Biological Sciences Outstanding MS Student (2023/2024), ISU
May 2024	Phi Sigma Publication Award (2023/2024), ISU
Oct 2022	Lela Winegarner Fellowship
Sept 2022	BirdFEEDER Research Grant Award
Sept 2022	College of Science Travel Grant, ISU
June 2022	SigmaXI Grants Aid of Research
May 2022	Mockford–Thompson Research Fellowship
May 2022	Robert D. Weigel Research Grant
Feb 2022	Phi Sigma Symposium – 2 nd Place Graduate Poster Presentation
May 2021	Mockford–Thompson Research Fellowship
May 2021	Robert D. Weigel Research Grant

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PRESENTATIONS

Sept. 2025	Gracious Donkor, Matthew Georgiou and James Slauch. Characterizing the regulatory interaction between HilD and HilE in <i>Salmonella</i> . <i>Midwest Microbial Pathogenesis Conference, Loyola, IL</i> . [Poster presentation]
March 2024	Donkor G. Investigating synthetic lethal interactions in <i>Vibrio cholerae</i> cell wall metabolism. <i>MCB Recruitment Weekend, UIUC, IL</i> [Poster presentation]
Sept. 2022	Donkor G., Stadler M., Anderson G., Dahl J. Investigating the antibiotic synergizing effects of the novel antimicrobial AGXX. <i>Midwest Microbial Pathogenesis Conference, Madison, WI</i> . [Poster presentation]
Aug. 2022	Donkor G., Stadler M., Anderson G., Dahl J. Investigating the antibiotic synergizing effects of the novel antimicrobial AGXX. <i>Molecular Genetics of Bacteria and Phages Meeting, Madison, WI</i> . [Poster presentation]
Nov. 2021	Donkor G., Anderson G., Hoffman K., Dahl J. Understanding <i>P. aeruginosa</i> 's response to the novel antimicrobial AGXX. <i>Midwest Microbial Pathogenesis Conference, East Lansing, MI</i> . [Poster presentation]

TEACHING & MENTORSHIP

UIUC: Taught discussion sessions for Molecular and Cellular Biology (MCB 250) and practical bench sessions for the laboratory complement of the course, MCB 251.

UIUC: 2025–Present: Mentoring two undergraduate students in the Slauch lab to develop an *in vivo* protein dimerization assay to investigate the interaction between a major transcriptional regulator of the *Salmonella* SPI1–T3SS and its principal post–translational negative regulator.

ISU: 2021–2022: Taught a total of 60 students in the laboratory sections for Microbiology and Society (BSC 160) and Microbiology (BSC 160). Trained students in basic bacterial culturing, pathogen identification techniques and molecular biology techniques such as PCR and Sanger sequencing.

ISU: 2021–2022: Trained two undergraduate students in the Dahl lab on biochemical and microbiological techniques to contribute to two research projects. Designed and supervised a summer workshop for over 15 high school students to screen soil microbial isolates for secondary metabolites active against *ESKAPE* pathogens.

PROFESSIONAL ORGANIZATIONS

Sigma Xi Scientific Research Honors Society	Mar 2026 – Present
American Society of Microbiology (ASM), Member	Aug 2021 – Present
Phi Sigma Biological Sciences Honors Society	Jan 2021 – Present
Convention for Biomedical Research Ghana (CoBReG)	Nov 2019 – Present

VOLUNTEERSHIP

Peer Reviewer for Microbiology Spectrum (ASM)	Feb 2026-present
Poster Judge for UIUC MCB Undergraduate Research Symposium	Sept 2023
MCB Graduate Student Association Student Member	Aug 2023 – Present
Business Manager, Phi Sigma Honors Society–ISU	Aug 2021–May 2023
Illinois Summer Research Academy, Microbiology Division – ISU	July 2021 & 2022
Social Chair, Phi Sigma Honors Society–ISU	Aug 2021–May 2022
Cofounder, Shared–Shelf Foundation, Ghana	June 2021–present
Community Health Screening Coordinator, Epithelium Diagnostics	April–Sept. 2018
Editor-in–Chief, Student Magazine, MELTSA–KNUST.	Sept 2015–May 2016